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Fig. 5.1 shows a circular coil of 500 turns and radius 0.12 m.

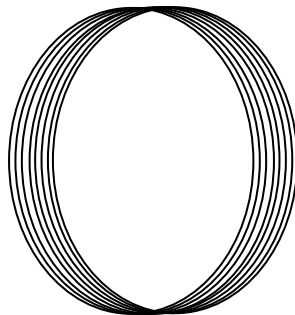


Fig. 5.1

A uniform magnetic field of flux density B is applied at right angles to the plane of the coil.

[Turn over

The magnetic flux density B changes with time t as shown in Fig. 5.2.

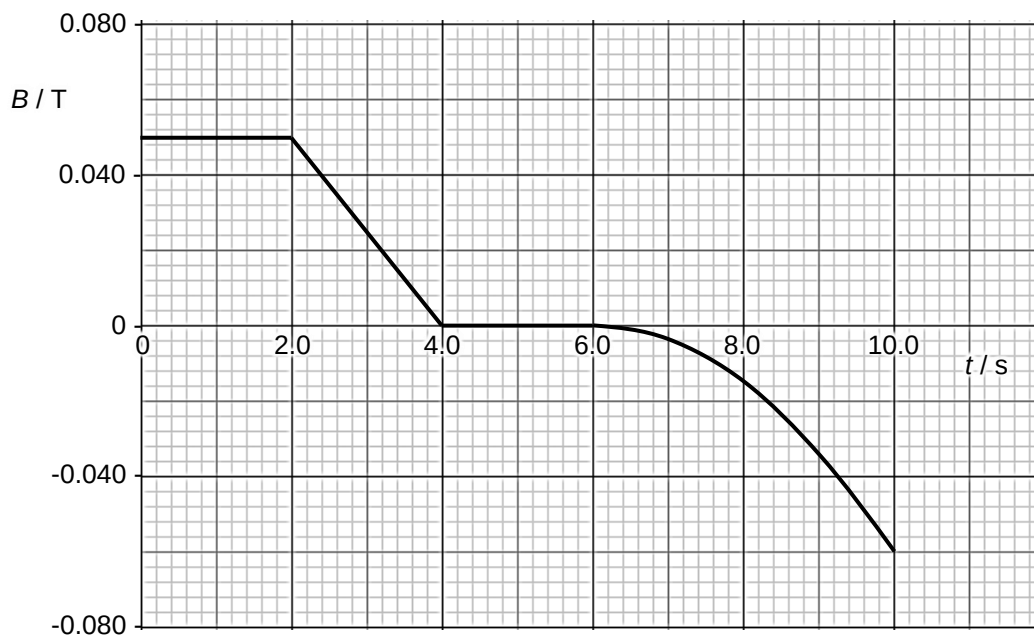


Fig. 5.2

From $t = 6.0$ s to $t = 10.0$ s, the gradient of the graph of B against t changes at a constant rate.

(a) Calculate the magnetic flux linkage of the coil at $t = 10.0$ s.

magnetic flux linkage = Wb [2]

L2 Magnetic flux linkage $\Phi = NBA$,
where A is the area and N is the number of the coil.
 $\Phi = NBA = (500)(0.060)\left[\pi(0.12^2)\right]$
 $= 1.3572 \text{ Wb} = 1.4 \text{ Wb (2 s.f.)}$

M1
A1

(b) Show that the magnitude of the induced e.m.f. in the coil between $t = 2.0$ s and $t = 4.0$ s is 0.57 V.

[2]

L2 Applying Faraday's law of electromagnetic induction,
 $|E| = \left| -\frac{d\Phi}{dt} \right| = \left| \frac{d(NBA)}{dt} \right| = NA \left| \frac{dB}{dt} \right|$ where $\frac{dB}{dt}$ is the gradient of the B-t graph
 $= (500)\left[\pi(0.12^2)\right] \left| \frac{(0.050 - 0)}{(2.0 - 4.0)} \right|$
 $= 0.56549 \text{ V} = 0.57 \text{ V (Shown)}$

C1
M1
A0

(c) On Fig. 5.3, sketch a graph to show the variation with time t of the induced e.m.f. E in the coil for time $t = 0$ to $t = 10.0$ s.

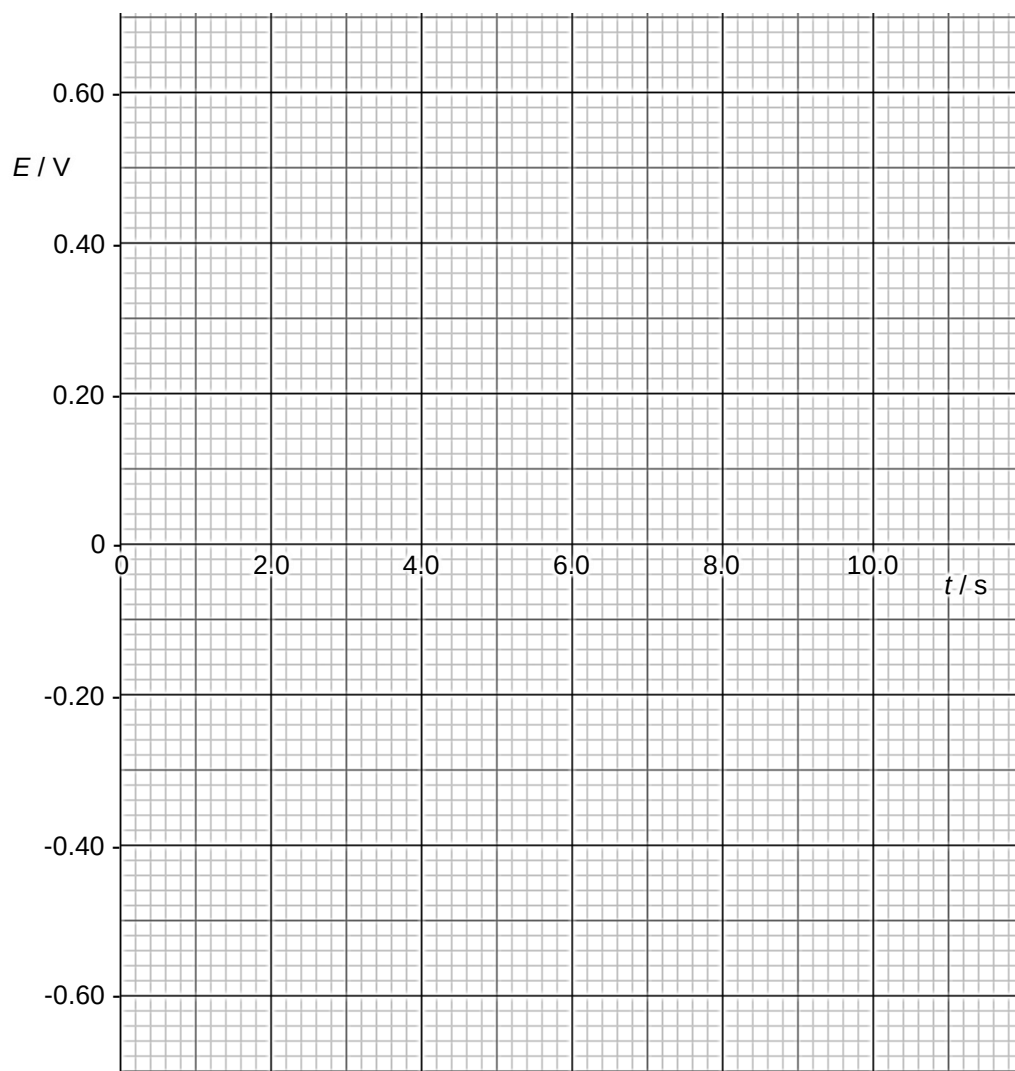
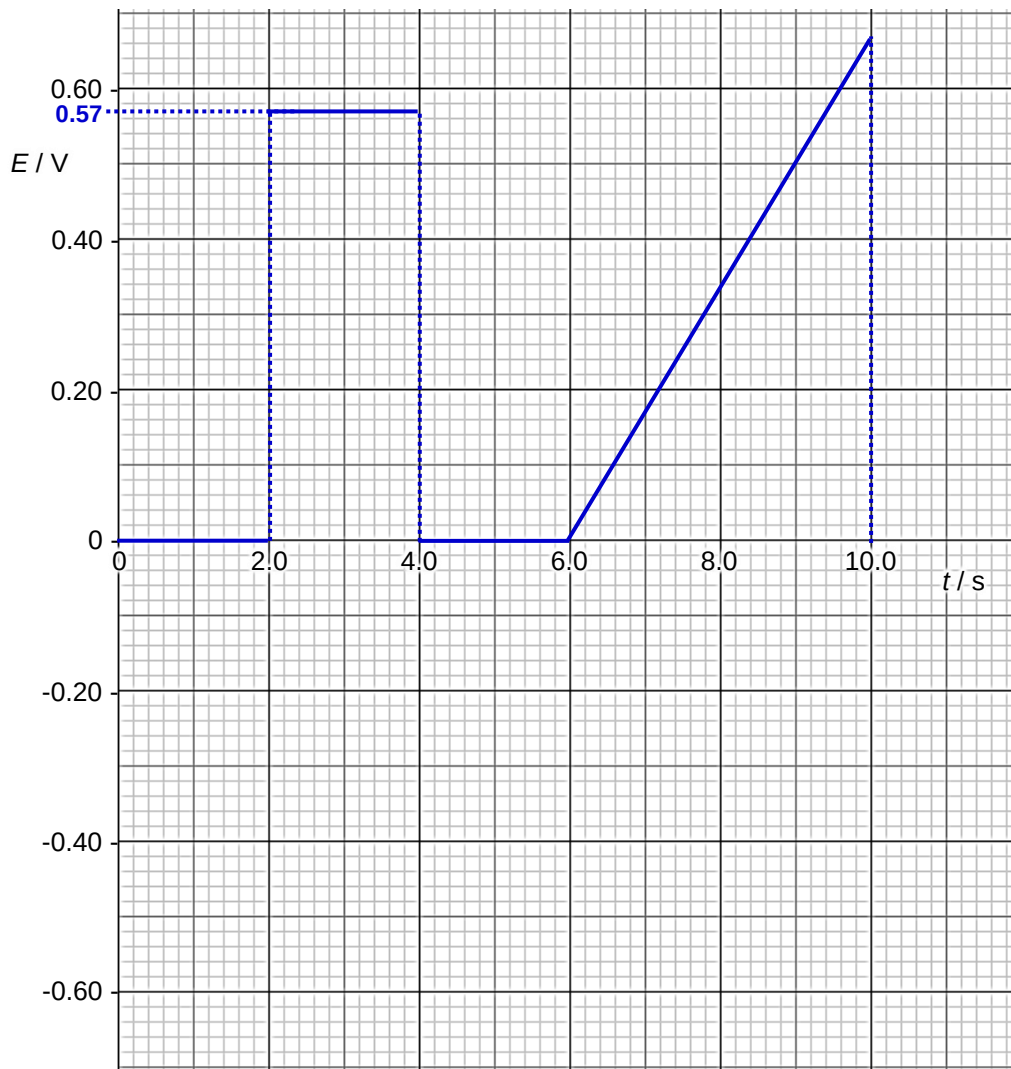


Fig. 5.3

[3]

L3

Solution:



From Faraday's Law and Lenz's Law,

$$E = -\frac{d\Phi}{dt} = -\frac{d(NBA)}{dt} = -NA \frac{dB}{dt}$$

Therefore,

1 mark:

Zero induced e.m.f. from 0 to 2.0 s and from 4.0 s to 6.0 s.

B1

1 mark:

Constant positive e.m.f. of 0.57 V from 2.0 s to 4.0 s.

B1

1 mark:

E.m.f. increases with time (straight line, positive gradient) from 6.0 s to 10.0 s.

B1

*Additional note to students (not marked for here): At 10.0 s, since the gradient dB/dt is **steeper** than the gradient from 2.0 s to 4.0 s, the magnitude of induced e.m.f. at 10.0 s is **greater than 0.57 V** (actual value not required since an accurate tangent cannot be constructed at the end of the graph in this case; if possible, the tangent should be drawn and gradient used to calculate the e.m.f. at 10.0s).*

*Note: There is **no reversal** of direction of induced e.m.f. at 6.0 s because the magnitude of B is increasing from 6.0 s to 10.0 s. So by Lenz's Law, in an attempt to oppose the change in flux linkage when B is reversed in direction and increasing in magnitude, the induced e.m.f. remains in the same direction as from 2.0 s to 4.0 s where B was of opposite direction and decreasing in magnitude.*

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[Total: 7]

6	A cylinder that contains a fixed amount of an ideal gas is shown in Fig. 6.1
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